

Kevin Babakhani

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EDUCATION

California State Polytechnic University, Pomona - B.S, Computer Science December 2024 - 3.83 GPA

Coursework: Software Engineering, Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, Mobile Application Development, Game Development, Artificial intelligence, Operating Systems

Awards and Involvement: Summa Cum Laude, Presidents List (2024), CPP Dean's List (2023-2024)

WORK EXPERIENCE

Coding Instructor - Coding Mind Academy - Remote October 2024 - Present

- Teach 10+ K-12 students a wide range of programming languages and technologies, including **Python, Java, JavaScript, HTML/CSS**, and **AI/Prompt Engineering**.
- Design and deliver interactive coding lessons tailored to different age groups and skill levels, achieving a high student engagement rate based on feedback and participation.
- Lead hands-on coding projects and workshops, helping students increase problem-solving proficiency as measured by pre- and post-assessments.
- Assess student progress and provide personalized feedback to 20+ learners, improving comprehension and retention by 25% based on student evaluations.

Operations Attendant - Associated Students Incorporated (CPP) - Pomona CA January 2023 - December 2024

- Managed an inventory system tracking 500+ student equipment checkouts, increasing efficiency and user satisfaction by 35% by improving organization and automating processes.
- Optimized task assignments and workflows by collaborating in team meetings, reducing operational bottlenecks and improving response times.
- Implemented and enforced safety protocols, responding to 3+ emergency incidents and ensuring 100% compliance with campus safety standards.

PROJECTS

SerenityStrings TypeScript/ReactJS/Python/FastAPI/TailwindCSS/ToneJS/Axios

- Developed Serenity Strings, a full-stack music theory and guitar learning platform featuring an interactive fretboard, chord progression generator, scale explorer, metronome, and guitar tuner to help musicians practice, compose, and improve their understanding of music theory.
- Built a responsive virtual guitar fretboard using **ReactJS, TypeScript**, and **TailwindCSS**, implementing dynamic note mapping, scale visualization, chord highlighting, alternate tunings, and mobile-friendly interfaces.
- Developed music theory algorithms and **FastAPI** backend services in **Python**, integrating **Axios** and **ToneJS** to support chord generation, scale construction, Roman numeral analysis, progression recommendations, real-time audio playback, metronome timing, and pitch-detection functionality for composition, improvisation, ear training, and instrument tuning.

LuminosityLEDs TypeScript/NextJS/ReactJS/NodeJS/ExpressJS/MongoDB/Axios

- Fullstack** Developer since October 2023 at LuminosityLEDs, designing and implementing a web app to control **embedded** LED lights using **Next.js** and **React components**.
- Developed and maintained **Dockerized microservices** for an **ActiveMQ**-based notification system, enabling reliable message queuing and real-time notifications in a **production cloud** environment.
- Optimized **MongoDB** connection handling across multiple **services**, optimizing efficiency.
- Collaborated in an **Agile** environment, working with **Kanban** and **Scrum** methodologies and participating in weekly team meetings to drive project progress effectively.

SerenityNest Flutter/Dart/Firebase

- Developed SerenityNest, a mental health **Android** app created with the **Flutter** framework and **Dart** language to support users' well-being through meditation, journaling, resources, and a quote-of-the-day page.
- Implemented secure **user authentication**, integrating email/password and Google sign-in, and **Firebase** for authentication and data storage.

SKILLS, HACKATHONS & ORGANIZATIONS

- Skills:**
 - Languages:**TypeScript, JavaScript, HTML/CSS, Java, C++, Python, Dart, C#, C
 - Technologies/Frameworks:** NextJS, React, Tailwind CSS, MongoDB, NodeJS, ExpressJS, Flutter, Mongoose, Firebase, MySQL, Axios, Docker, Vite, Blender, Unity, Vercel
 - Developer Tools:** Git/GitHub, Atlassian: Confluence/Jira, Figma, Notion, Linux
- Hackathons:** CPP 2024 BroncoHacks, CPP 2025 BroncoHacks
- Organizations:** 2025 T. Howard Foundation Final Talent Pool Member